Global–Local Coupled Style Transfer for Semantic Segmentation of Bitemporal Remote Sensing Images

Hao Wang, Mingning Guo, Shaoxian Li, Haifeng Li, *Member, IEEE*, and Chao Tao, *Member, IEEE*

***Abstract*— Due to the different acquisition conditions, large variations in the feature distributions of two temporal domains generally exist, known as temporal domain shift. The temporal domain shift is primarily influenced by coupled dual-factor: global style variations (such as illumination and weather conditions) and local style variations (such as the inherent phenological properties of land cover classes). In this article, we first formulate the temporal domain shift problem as an issue of dual-factor coupled interference on feature distributions in remote sensing (RS) community. To address this issue, we propose a semantic-guided style transfer (SGST) framework seamlessly integrating global feature alignment with local feature semantic matching. We use an adaptive segmentation model to provide pseudosegmentation maps and feed them into the style transfer model as semantic guidance. Under semantic guidance, a semantic-constrained style normalization (SCSN) module is designed to achieve style transfer at both global and local levels. Furthermore, a dual learning approach is introduced to make the style transfer model and the adaptive segmentation model promote each other. As a result, the style transfer model generates high-quality style-transferred images, and the adaptive segmentation model progressively predicts more accurate pseudosegmentation maps. Extensive experiments demonstrate the superiority of our proposed framework over state-of-the-art methods in terms of both perceptual quality and quantitative performance.**

***Index Terms*— Dual learning, global–local coupled style transfer, semantic guidance, temporal domain shift.**

I. Introduction

With the development of sensor technology, the frequency of remote sensing (RS) image acquisition is rapidly increasing. This poses higher requirements for temporal RS image processing and analysis. Semantic segmentation of bitemporal RS images is one of the most fundamental and important tasks for updating land cover maps, which has a wide range of applications in land cover monitoring [35], [46], ecological conservation [5], [14], agricultural production [1], [16], and other fields. However, RS images acquired from different times or seasons often exhibit significant visual differences [26], [34], resulting in a significant performance drop when models trained on labeled data from the past time (source domain) are applied to unlabeled data from the current new time (target domain). We refer to this phenomenon as the temporal domain shift in RS images.

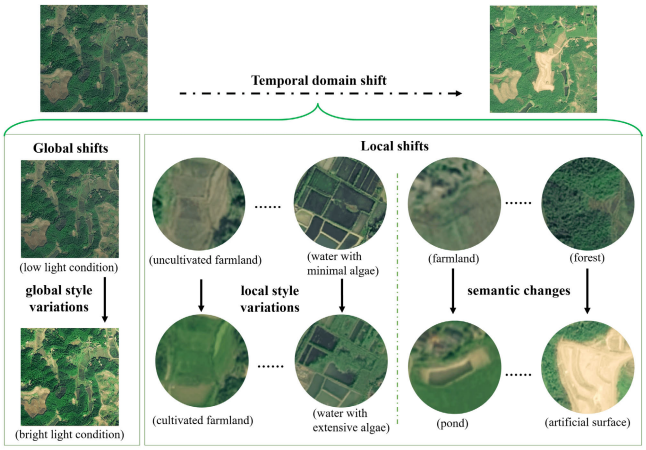


Fig. 1. Illustration of the temporal domain shift influenced by: 1) global shifts induced by global style variations and 2) local shifts caused by semantic changes and local style variations. Semantic changes display significant randomness and lack stable temporal patterns. It is worth noting that both global and local style variations exhibit periodic temporal patterns.

As shown in Fig. 1, the temporal domain shift in RS images is influenced by two dominant factors. First, global shifts induced by global style variations. Global style variations (such as illumination, weather conditions, and viewing angles) uniformly affect the imaging of all land cover classes [30], resulting in global feature differences between bitemporal RS images. Second, local shifts caused by semantic changes and local style variations. Semantic changes, often triggered by human activities [49] or natural disasters [6], display significant randomness and lack stable temporal patterns. In contrast, local style variations, which are induced by the inherent phenological properties of land cover classes, exhibit periodic temporal patterns. Phenology refers to periodic variations in the adaptation of land cover classes to seasonal climate conditions, and it differentially affects the imaging of each land cover class, manifesting as local feature differences between bitemporal RS images [3], [38]. It is worth noting that different land cover classes exhibit phenological heterogeneity, which can be observed in two aspects. First, in contrast to

Manuscript received 18 December 2023; revised 27 April 2024 and 1 June 2024; accepted 5 July 2024. Date of publication 9 July 2024; date of current version 19 July 2024. This work was supported in part by the Natural Science Foundation of Hunan for Distinguished Young Scholars under Grant 2022JJ10072, in part by the National Natural Science Foundation of China under Grant 42171376 and Grant 41771458, in part by the Open Project of Xiangjiang Laboratory under Grant 22XJ03007, and in part by the High-Performance Computing Center of Central South University. *(Corresponding author: Chao Tao.)*

The authors are with the School of Geosciences and Info-Physics, Central South University, Changsha 410083, China (e-mail: kingtaochao@csu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3425672

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

nonvegetated areas, vegetation is more susceptible to undergo visual differences with seasonal changes, such as plant sprouting, leaf growth, leaf color change, and leaf shedding. [4]. Second, various types of vegetation follow distinct phenological rules. For example, broadleaf evergreen forests exhibit varying degrees of greenness throughout the seasons, while farmlands may be filled with green crops in summer but in a fallow period in winter [39]. Therefore, phenology-induced local style variations can lead significant feature differences of land cover types in the same geographic location, thus challenging semantic segmentation models under cross-temporal domain condition. If there were no local style variations, it may become a problem of intradomain semantic segmentation even exists local semantic changes between bitemporal images in the same geographic location. Consequently, the impact of local style variations on cross-temporal domain semantic segmentation is significantly larger than local semantic changes. Based on the above analysis, we focus on developing models to mitigate the effects of both global and local style variations on cross-temporal domain semantic segmentation.

To mitigate the domain shift, domain adaptation (DA) has been presented to improve the performance of cross-domain semantic segmentation models. We classify techniques for DA of semantic segmentation into two categories: global adaptation and local adaptation.

Global adaptation learning aims to learn domain-invariant features between source and target domains. One common solution is global adversarial adaptation, which trains a discriminator to distinguish features [8], [12], [32], [48] or segmentation outputs [29], [33], [41] from the source and target domain while training a classifier to fool the discriminator and extract domain-invariant features. In the RS field, several approaches [2], [40], [43] have been proposed to improve the discriminator's ability to learn a domain-invariant classifier for RS image semantic segmentation. Another solution is global style transfer. The main idea of global style transfer is to transfer the global visual appearance of images from source domain to target domain through some generative constraints, such as cycle-consistency loss [15], [22], [44], [51], instance normalization (IN) [31], or adaptive IN [13], making transferred images have similar feature distributions to target domain. Then, segmentation model can be trained on both original source images and transferred images to learn domain-invariant features. In the RS community, they design style transfer methods to reduce the differences among RS images taken by different sensors [19], [24], composed of varying spectral bands [20], or acquired under various atmospheric conditions [10], [27]. However, these methods focus on reducing global distribution differences, neglecting local differences caused by inherent phenological properties of the land cover class. In reality, global alignment is a random mapping within a large feasible solution space. When different land cover classes in the target domain have heterogeneity in distribution alignment, global alignment may lead to excessive alignment of features from different land cover classes, resulting in feature mismatch.

Local adaptation learning aims to learn adaptive features according to local correlations between source and target domains. One common solution is self-learning. The core idea of self-learning is to train target segmentation model through target pseudolabels. These pseudolabels are predicted with high confidence by the trained source model and contain some associated information between source and target domains. They learn adapted model by enhancing the quality of pseudolabels, such as confidence selection [17], [52], confidence regularization [53], prototypical denoising [47], mixed sampling [28], or introducing local spatial priors [21], [25]. Another solution is local adversarial adaptation. They design specific adversarial loss to independently align the feature distribution of different local regions, based on consistency regularization [18], [23], [36] or local region priors [37], [45]. Recently, in the RS field, some works adopt self-learning combined with adversarial learning [42], [50] to reduce feature distribution differences among cross-domain RS images. However, these methods focus on feature adaptation among local regions while giving little consideration to semantic correlations of local regions within the global context, which tends to suffer from the inconsistency of semantic features. For example, paddy fields and dry fields, despite belonging to the farmlands category, typically exhibit significant differences in RS images. These local adaptation methods might align features within paddy fields or dry fields but increase the feature differences between these two types of farmlands, ultimately resulting in the inconsistency of semantic features.

To address the aforementioned challenges, we propose a semantic-guided style transfer (SGST) framework seamlessly integrating global feature alignment with local feature semantic matching. SGST combines a style transfer model and an adaptive segmentation model into a dual learning manner. First, we employ the adaptive segmentation model to provide pseudosegmentation maps and feed them into the style transfer model as semantic guidance. Subsequently, under semantic guidance, a semantic-constrained style normalization (SCSN) module is designed to achieve style transfer at both global and local levels. Furthermore, we iteratively optimize the style transfer model and the adaptive segmentation model to make them promote each other. On the one hand, with the guidance of pseudosegmentation maps, the style transfer model could generate high-quality style-transferred images. On the other hand, with the boost of improved style-transferred images, the adaptive segmentation model progressively predicts more accurate pseudosegmentation maps. In summary, the main contributions of this article can be summarized as follows.

1) This article first formulates the temporal domain shift problem as an issue of dual-factor coupled interference on feature distributions in RS community. To effectively address this issue, we propose a SGST framework that seamlessly integrates global feature alignment with local feature semantic matching.
2) We design a SCSN module that enables style transfer at both global and local levels. This results in style-transferred images that exhibit a similar visual appearance within the same land cover class while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

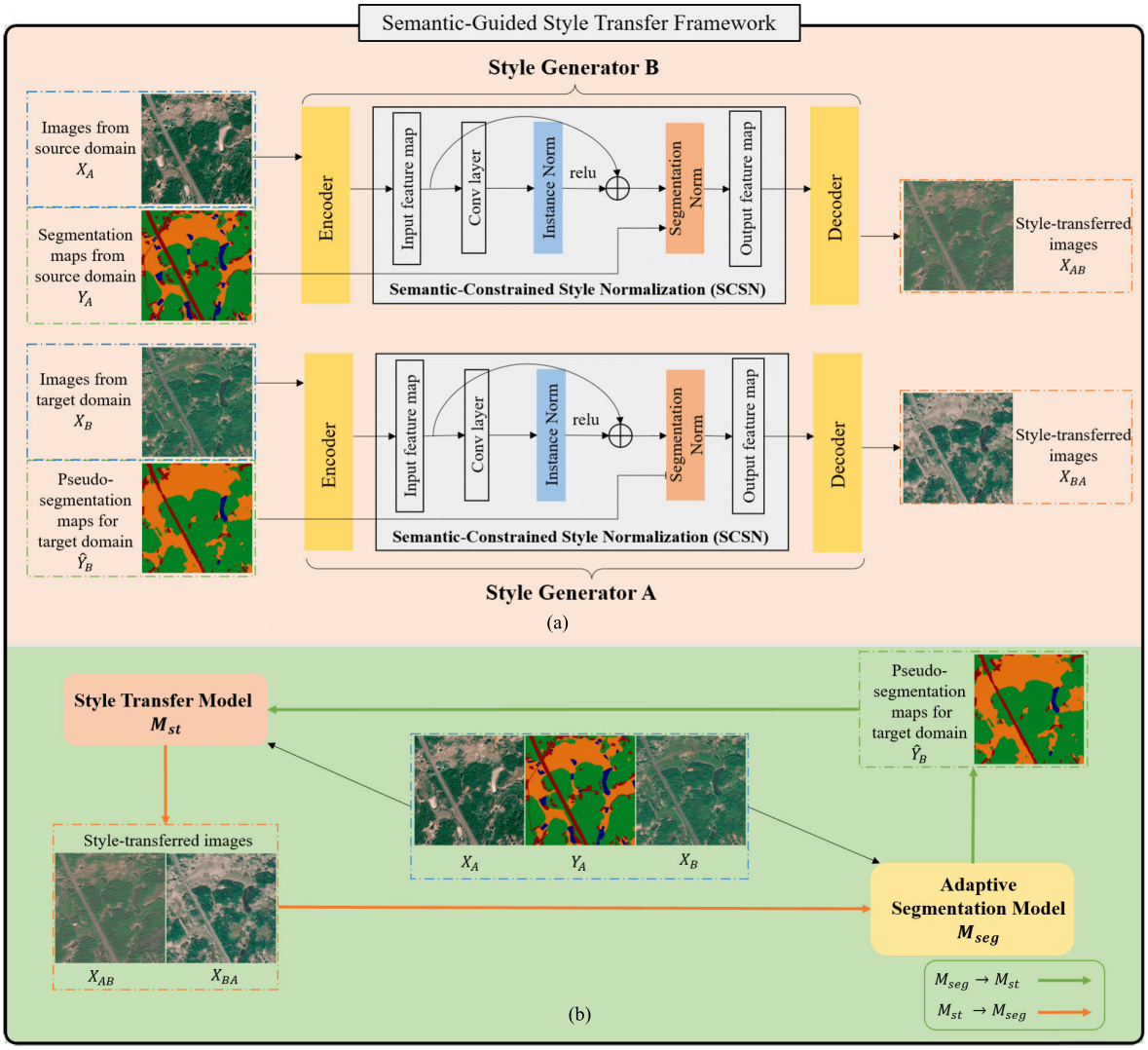


Fig. 2. Overview framework. Under semantic guidance, the SCSN module is proposed to enable style transfer at both global and local levels. The dual learning approach is introduced to make the style transfer model and the adaptive segmentation model promote each other. (a) Global–local coupled style transfer via semantic guidance. (b) Dual learning between style transfer model and adaptive segmentation model.

preserving heterogeneity across different land cover classes. A dual learning approach is also introduced to make the style transfer model and the adaptive segmentation model promote each other.

3) We validate our method on three real-world bitemporal RS image datasets. We conduct compared experiments against other DA methods. We also conduct ablation experiments to analyze the characteristics of our method in detail. The experimental results demonstrate that the proposed framework significantly enhances the performance in DA task of semantic segmentation for bitemporal RS images.

II. Method

A. Overview Framework

Given the labeled source, temporal dataset X_A with associated K-class segmentation maps Y_A and the unlabeled target temporal dataset X_B. Our goal is to minimize temporal domain shift as much as possible to make the knowledge learned in the source temporal domain transfer to the target temporal domain.

We formulate the temporal domain shift primarily influenced by coupled dual-factor: global style variations and local style variations. For the style transfer from X_A to X_B, global style transfer function can be denoted as $h(X_A)$, which uniformly affects the imaging of all land cover classes. Local style transfer function can be denoted as $h_k(Y_A^k \cdot X_A)$, which differentially affects the imaging of each land cover class k by applying a semantic mask Y_A^k. Hence, the coupled style transfer from X_A to X_B can be denoted as

$$h(X_A) + \sum_{k=1}^{K} h_k\left(Y_A^k \cdot X_A\right). \tag{1}$$

When target pseudosegmentation maps $\hat{Y}_B$ are available, we can also denote the coupled style transfer from X_B to X_A as

$$h(X_B) + \sum_{k=1}^{K} h_k\left(\hat{Y}_B^k \cdot X_B\right). \tag{2}$$

Fig. 2 gives an overview of our proposed framework, which combines a style transfer model M_{st} and an adaptive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

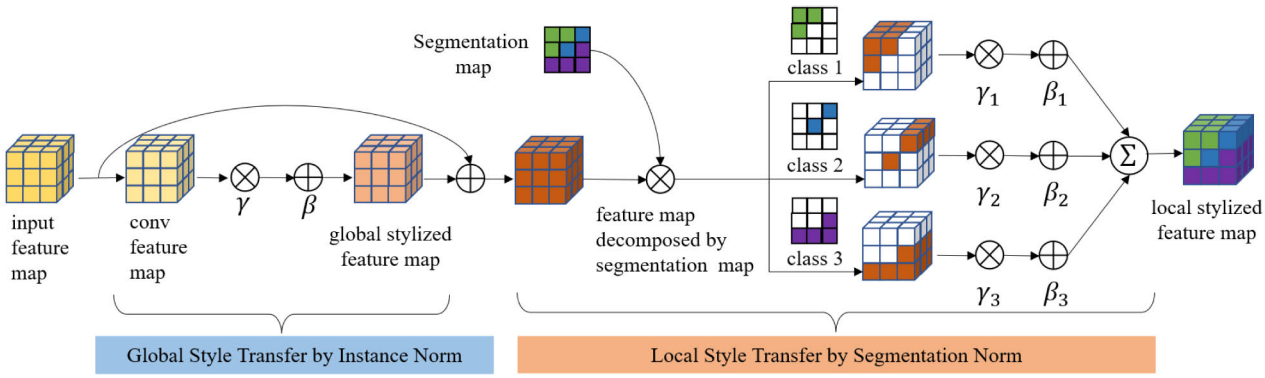


Fig. 3. Illustration of IN and SN.

segmentation model M_{seg} into a dual learning manner. To make style transfer at both global and local levels, we propose a SCSN module that takes source segmentation maps Y_A and target pseudosegmentation maps $\hat{Y}_B$ generated from the adaptive segmentation model as semantic guidance in the style transfer process. To make the style transfer model and the adaptive segmentation model promote each other, we iteratively optimize the style transfer model and the adaptive segmentation model.

In the following sections, Section II-B provides a detailed introduction to the SCSN module, the architecture of style transfer network, and loss functions. In Section II-C, we present a comprehensive description of the dual learning between the style transfer model and the adaptive segmentation model.

B. Global–Local Coupled Style Transfer via Semantic Guidance

1) Semantic-Constrained Style Normalization: IN [9], [13], [31] is a general normalization method for global style transfer. However, IN essentially only imposes stylized constraints on global features of an image, allowing it to pay more attention to the global style but falling short in capturing the local style. Therefore, we propose a SCSN module that enables style transfer at both global and local levels. This module utilizes source segmentation maps and target pseudosegmentation maps as semantic guidance. With the help of SCSN, the style transfer model generates high-quality style-transferred images that exhibit a similar visual appearance within the same land cover class while preserving heterogeneity across different land cover classes.

For the source domain, given an input batch $x_A \in \mathbb{R}^{(N\times C\times H\times W)}$, along with the corresponding one-hot segmentation maps $y_A \in \mathbb{R}^{(N\times K\times H\times W)}$, where N, C, and K represent the batch size, feature channels, and class numbers. We denote the final activation of the encoder from source data as f_A. The SCSN module comprises an IN layer and a segmentation normalization (SN) layer, which can be formulated as

$$\mathcal{G}(f_A) = \text{IN}(\mathcal{F}(f_A)) + f_A \tag{3}$$

$$z_A = \text{SN}(\mathcal{G}(f_A), y_A) \tag{4}$$

where z_A represents the output stylized feature maps from SCSN. The function $\mathcal{F}(\cdot)$ denotes the convolutional layer, and the function $\mathcal{G}(\cdot)$ denotes the residual connection [11] with IN. The operation $\text{SN}(\mathcal{G}(f_A), y_A)$ takes global stylized feature maps and segmentation maps as input to achieve local semantic stylized. As shown in Fig. 3, the main difference from IN is that SN does not directly perform stylized on whole feature maps. Instead, it decomposes whole feature maps into several local semantic feature maps, and then applies normalization to each local semantic feature map before summing them up. In summary, to achieve style transfer at both global and local levels, SCSN first performs global stylized by adjusting global statistical properties of the whole feature maps and then conducts refined local stylized by adjusting local statistical properties of each semantic feature map. It should be noted that the spatial scales of $\mathcal{G}(f_A)$ and y_A must be equal. If their spatial scales are not equal (e.g. spatial scales of feature maps may decrease after convolution), we can resample y_A to guarantee equal spatial scales between $\mathcal{G}(f_A)$ and y_A

$$z_A = \text{SN}(\mathcal{G}(f_A), r(y_A)) \tag{5}$$

here, $r(\cdot)$ represents the resampling function.

Specifically, IN performs mean and variance normalization on each feature channel of each instance sample

$$\text{IN}(f_A) = \gamma\left(\frac{f_A - \mu_{nc}(f_A)}{\sigma_{nc}(f_A)}\right) + \beta \tag{6}$$

$$\mu_{nc}(f_A) = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W} f_A \tag{7}$$

$$\sigma_{nc}(f_A) = \sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}(f_A - \mu_{nc}(f_A))^2 + \epsilon} \tag{8}$$

where γ and $\beta \in \mathbb{R}^C$ are the learned affine parameters from the data. Under semantic guidance, we can obtain the local semantic feature for class k by applying a semantic mask y_A^k, resulting in $(y_A^k f_A) \in \mathbb{R}^{(N\times C\times H\times W)}$. In this operation, local positions that correspond to classes other than k are replaced with zeros. SN performs channel normalization on each local semantic feature map from an instance sample

$$\text{SN}(f_A, y_A) = \sum_k y_A^k\left(\gamma^k \frac{y_A^k f_A - \mu_{nc}^k(f_A, y_A)}{\sigma_{nc}^k(f_A, y_A)} + \beta^k\right) \quad k \in [1, K] \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

$$\mu_{nc}^{k}(f_A, y_A) = \frac{\sum_{h=1}^{H}\sum_{w=1}^{W}\left(y_A^k f_A\right)}{\sum_{h=1}^{H}\sum_{w=1}^{W} y_A^k} \tag{10}$$

$$\sigma_{nc}^{k}(f_A, y_A) = \sqrt{\frac{\sum_{h=1}^{H}\sum_{w=1}^{W}\left(y_A^k f_A - \mu_{nc}^{k}(f_A, y_A)\right)^2}{\sum_{h=1}^{H}\sum_{w=1}^{W} y_A^k} + \epsilon} \tag{11}$$

where γ and $\beta \in \mathbb{R}^{K\times C}$ are the learned affine parameters corresponding to land cover classes from the data.

For the target domain, given an input batch $x_B \in \mathbb{R}^{(N\times C\times H\times W)}$, we employ an adaptive segmentation model to generate pseudosegmentation maps $\hat{y}_B \in \mathbb{R}^{(N\times K\times H\times W)}$ as semantic guidance for style transfer. The adaptive segmentation model uses DeepLab v3+ [7] with ResNet-50 and is initially trained on labeled source dataset. We denote the final activation of the encoder from target data as f_B. Correspondingly, replacing the source input batch in (4) and (5) with the target input batch, the SCSN module can be formulated as

$$\mathcal{G}(f_B) = \mathrm{IN}(\mathcal{F}(f_B)) + f_B \tag{12}$$

$$z_B = \mathrm{SN}(\mathcal{G}(f_B), r(\hat{y}_B)). \tag{13}$$

2) Style Transfer Network and Loss Functions: We impose the proposed SCSN module into style transfer network to generate high-quality style-transferred images that exhibit a similar visual appearance within the same land cover class while preserving heterogeneity across different land cover classes. We denote G_A and G_B as the style generator for source domain and target domain, respectively. A style generator consists of an encoder, a SCSN module, and a decoder. We denote θ_{G_A} and θ_{G_B} as the learnable parameters in G_A and G_B, respectively. Given the corresponding one-hot segmentation maps Y_A for source images X_A. Given pseudosegmentation maps $\hat{Y}_B$, generated by the adaptive segmentation model for target images X_B. We combine X_A and Y_A with θ_{G_B} to generate transferred images X_{AB} which are stylized as domain B. Similarly, we combine X_B and $\hat{Y}_B$ with θ_{G_A} to generate transferred images X_{BA} which are stylized as domain A.

We employ an adversarial strategy to optimize the style transfer model by fooling a discriminator. The discriminator consists of several shared layers and multiple domain-specific layers. The discriminator takes both style-transferred images and real images as input. The output is a probability value ranging from 0 to 1, where a value closer to 1 indicates a higher degree of style authenticity. D_A and D_B represent the discriminator's outputs for domain A and B, respectively. The loss functions of the discriminator are defined as follows:

$$\mathcal{L}_{D_{\mathrm{adv}}}(A, B) = -\left[\mathbb{E}_{x_A}[\log(D_A(x_A))] + \mathbb{E}_{x_{BA}}[\log(1 - D_A(x_{BA}))]\right] \tag{14}$$

$$\mathcal{L}_{D_{\mathrm{adv}}}(B, A) = -\left[\mathbb{E}_{x_B}[\log(D_B(x_B))] + \mathbb{E}_{x_{AB}}[\log(1 - D_B(x_{AB}))]\right] \tag{15}$$

$$\mathcal{L}_{D_{\mathrm{adv}}} = \mathcal{L}_{D_{\mathrm{adv}}}(A, B) + \mathcal{L}_{D_{\mathrm{adv}}}(B, A). \tag{16}$$

Correspondingly, adversarial loss functions for the style generators are defined as follows:

$$\mathcal{L}_{G_{\mathrm{adv}}}(A, B) = -\mathbb{E}_{x_{BA}}[\log(D_A(x_{BA}))] \tag{17}$$

$$\mathcal{L}_{G_{\mathrm{adv}}}(B, A) = -\mathbb{E}_{x_{AB}}[\log(D_B(x_{AB}))] \tag{18}$$

$$\mathcal{L}_{G_{\mathrm{adv}}} = \mathcal{L}_{G_{\mathrm{adv}}}(A, B) + \mathcal{L}_{G_{\mathrm{adv}}}(B, A). \tag{19}$$

Additionally, it is important to maintain content consistency between X_A and X_{AB}, as well as between X_B and X_{BA}. First, by combining X_{AB}, Y_A with θ_{G_A}, and X_{BA}, $\hat{Y}_B$ with θ_{G_B}, we generate cycle-transferred samples X_{ABA} and X_{BAB}. We minimize the cycle-consistency loss

$$\mathcal{L}_{\mathrm{cycle}} = |x_A - x_{ABA}| + |x_B - x_{BAB}|. \tag{20}$$

Finally, we combine X_A and Y_A with θ_{G_A} and X_B and $\hat{Y}_B$ with θ_{G_B} to generate reconstructed images X_{AA} and X_{BB}, which should also remain consistent content with the original images. We minimize the reconstruction consistency loss

$$\mathcal{L}_{\mathrm{recons}} = |x_A - x_{AA}| + |x_B - x_{BB}|. \tag{21}$$

Therefore, the overall objective function of style generators is defined as follows:

$$\mathcal{L}_G = \mathcal{L}_{G_{\mathrm{adv}}} + \mathcal{L}_{\mathrm{cycle}} + \mathcal{L}_{\mathrm{recons}}. \tag{22}$$

The data flow during the style transfer training is shown in Fig. 4.

C. Dual Learning Between Style Transfer Model and Adaptive Segmentation Model

Since pseudosegmentation maps generated by the adaptive segmentation model are initially coarse, the style transfer with semantic guidance might be affected. For this reason, we iteratively optimize the style transfer model M_{st} and the adaptive segmentation model M_{seg} in a dual learning manner to make M_{st} and M_{seg} promote each other. The proposed dual learning includes two optimization directions: ($M_{\mathrm{seg}} \rightarrow M_{\mathrm{st}}$) and ($M_{\mathrm{st}} \rightarrow M_{\mathrm{seg}}$). ($M_{\mathrm{seg}} \rightarrow M_{\mathrm{st}}$) denotes that we employ M_{seg} to generate pseudosegmentation maps as semantic guidance for the optimization of M_{st}. ($M_{\mathrm{st}} \rightarrow M_{\mathrm{seg}}$) denotes that we employ M_{st} to generate style-transferred images to boost the performance of M_{seg}. ($M_{\mathrm{seg}}^{(m-1)} \rightarrow M_{\mathrm{st}}^{(m)} \rightarrow M_{\mathrm{seg}}^{(m)}$) represents the m iteration of dual learning. The detailed optimization process is illustrated in Algorithm 1. $M_{\mathrm{seg}}^{(0)}$ represents the initial segmentation model trained solely on images and segmentation maps from the source domain, and $\hat{Y}_B^{(0)}$ denotes pseudo segmentation maps generated by $M_{\mathrm{seg}}^{(0)}$ for the target domain.

1) ($M_{\mathrm{seg}} \rightarrow M_{\mathrm{st}}$): By fixing the parameters of M_{seg}, we employ M_{seg} to generate pseudosegmentation maps for the target domain as semantic guidance for style transfer. Given probability prediction results of the target domain, denoted as $p_B = M_{\mathrm{seg}}(x_B)$, we apply a confidence threshold d to filter p_B, selecting highly confident predictions as pseudo labels. For a target pixel $x_B^{(i)}$, its one-hot pseudolabel $\hat{y}_B^{(i,k)}$ is expressed as

$$\hat{y}_B^{(i,k)} = \begin{cases} 1, & \text{if } k = \mathrm{argmax}_{k'}\left(p_B^{(i,k')}\right) \text{ and } p_B^{(i,k)} > d \\ 0, & \text{otherwise.} \end{cases} \tag{23}$$

Under semantic guidance of pseudosegmentation maps, M_{st} with the proposed SCSN module could generate improved style-transferred images that exhibit a similar visual appearance within the same land cover class while preserving heterogeneity across different land cover classes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

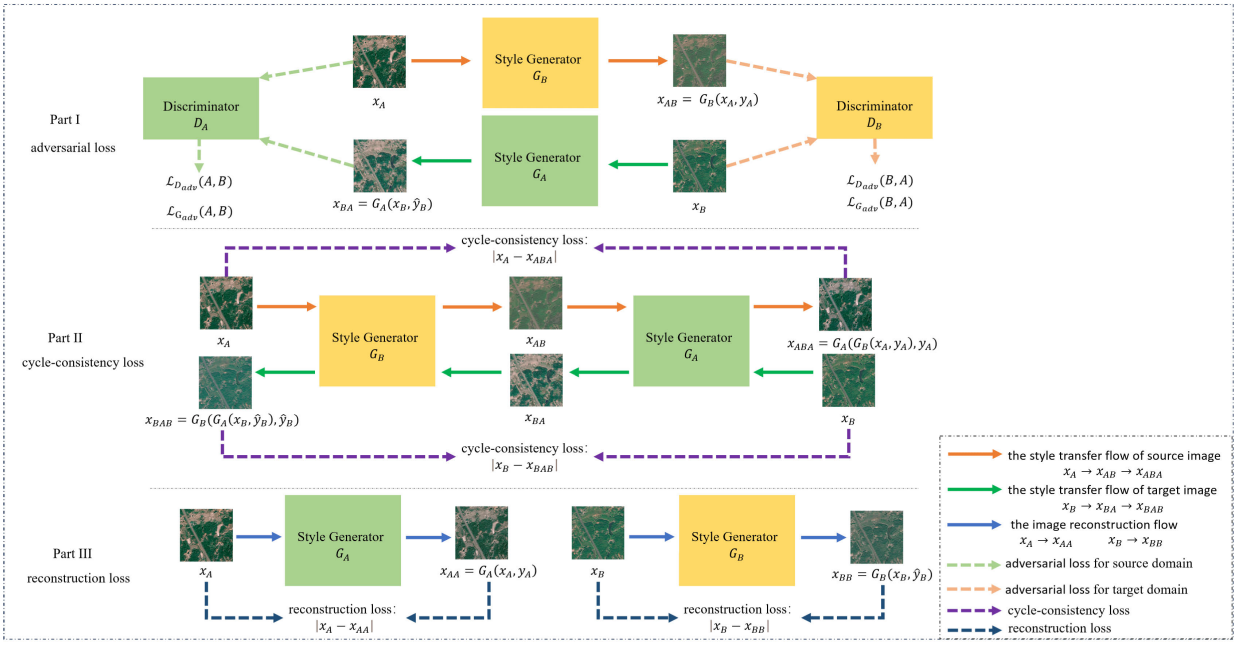


Fig. 4. Data flow during the style transfer training.

Algorithm 1 Dual Optimization Process Between Style Transfer Model and Adaptive Segmentation Model

Input: (X_A, Y_A), $(X_B, \hat{Y}_B^{(0)})$, $M_{seg}^{(0)}$
for $m \leftarrow 1$ *to* N **do**
  # Use $\hat{Y}_B^{(m-1)}$ as semantic guidance for the optimization of M_{st} #
  for $i \leftarrow 1$ *to* N_{st} **do**
    Sampling from (X_A, Y_A);
    Sampling from $(X_B, \hat{Y}_B^{(m-1)})$;
    Update $M_{st}^{(m)} \leftarrow M_{st}^{(m-1)}$ with Eq. 22, Eq. 16;
  end for
  # Generate style-transferred images over the source and target datasets #
  Generate $X_{AB}^{(m)} = M_{st}^{(m)}(X_A, Y_A, \theta_{G_B}^*)$
  Generate $X_{BA}^{(m)} = M_{st}^{(m)}(X_B, \hat{Y}_B^{(m-1)}, \theta_{G_A}^*)$
  # Use $X_{AB}^{(m)}$ and $X_{BA}^{(m)}$ to boost the performance of M_{seg} #
  for $j \leftarrow 1$ *to* N_{seg} **do**
    Sampling from $(X_A, Y_A, X_{AB}^{(m)})$
    Sampling from $(X_B, \hat{Y}_B^{(m-1)}, X_{BA}^{(m)})$
    Update $M_{seg}^{(m)} \leftarrow M_{seg}^{(m-1)}$ with Eq. 28
  end for
  # Update all pseudosegmentation maps over the target dataset #
  Update $\hat{Y}_B^{(m)} \leftarrow \hat{Y}_B^{(m-1)}$ with Eq. 23
end for
Output: $M_{seg}^{(m)}(M_{st}^{(m)})$

2) $(M_{\text{st}} \to M_{\text{seg}})$*:* During the initial optimization, $M_{\text{seg}}^{(0)}$ is trained on images and segmentation maps from the source domain. After $(M_{\text{seg}} \to M_{\text{st}})$, style-transferred images generated by M_{st} can be utilized to facilitate the adaptation learning. With the boost of style-transferred images, we optimize M_{seg} through self-learning loss and prediction consistency loss to progressively predict more accurate pseudosegmentation maps.

Specifically, given source probability predictions $p_A = M_{\text{seg}}(x_A)$ and one-hot segmentation maps y_A, the segmentation loss function of the source domain is defined as

$$\mathcal{L}_{\text{seg}} = -\sum_{i=1}^{H\times W}\sum_{k=1}^{K} y_A^{(i,k)} \log\left(p_A^{(i,k)}\right). \tag{24}$$

Given target probability predictions $p_B = M_{\text{seg}}(x_B)$ and generated pseudo segmentation maps $\hat{y}_B$ from the last loop, we define the self-learning loss function of the target domain as

$$\mathcal{L}_{\text{self}} = -\sum_{i=1}^{H\times W}\sum_{k=1}^{K} \hat{y}_B^{(i,k)} \log\left(p_B^{(i,k)}\right). \tag{25}$$

Given style transfer results (source-to-target) $x_{AB} = M_{\text{st}}(x_A, y_A, \theta_{G_B}^*)$, we obtain $p_{AB} = M_{\text{seg}}(x_{AB})$. For the high-confidence prediction regions in p_{AB}, they should maintain consistency with predictions on the real source images. Therefore, we minimize the source prediction consistency loss function

$$\mathcal{L}_{\text{cons}-s} = -\sum_{i=1}^{H\times W}\sum_{k=1}^{K} r_{AB}^{(i)} \left| p_{AB}^{(i,k)} - p_A^{(i,k)} \right| \tag{26}$$

where $r_{AB}^{(i)} = \begin{cases} 1, & \text{if } \max_k p_{AB}^{(i,k)} > d \\ 0, & \text{otherwise} \end{cases}$ represents the mask region of high-confidence predictions in $p_{AB}^{(i)}$.

Similarly, given style transfer results (target-to-source) $x_{BA} = M_{\text{st}}(x_B, \hat{y}_B, \theta_{G_A}^*)$, we obtain $p_{BA} = M_{\text{seg}}(x_{BA})$. For the high-confidence prediction regions in p_{BA}, they should also maintain consistency with predictions on the real target

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

images. Therefore, we minimize the target prediction consistency loss function

$$\mathcal{L}_{\text{cons}-t} = -\sum_{i=1}^{H\times W}\sum_{k=1}^{K} r_{BA}^{(i)}\left|p_{BA}^{(i,k)} - p_{B}^{(i,k)}\right| \tag{27}$$

where $r_{BA}^{(i)} = \begin{matrix}1, & \text{if } \max_k p_{BA}^{(i,k)} > d\\ 0, & \text{otherwise}\end{matrix}$ represents the mask region of high-confidence predictions in $p_{BA}^{(i)}$.

Therefore, the objective function of the adaptive segmentation model is defined as

$$\mathcal{L}_{\text{adapt}} = \lambda_1\mathcal{L}_{\text{seg}} + \lambda_2\mathcal{L}_{\text{self}} + \lambda_3\mathcal{L}_{\text{cons}-s} + \lambda_4\mathcal{L}_{\text{cons}-t}. \tag{28}$$

III. Experiments and Analysis

To demonstrate the domain adaptability of our SGST framework, we design a series of temporal cross-domain semantic segmentation experiments. In Section III-A, we first describe the experimental datasets and the details of implementation settings. In Section III-B, we conduct comparison experiments with state-of-the-art domain adaptive semantic segmentation methods.

A. Experimental Settings

1) Datasets: Datasets used in the experiments are acquired from Gaofen-2 satellite, with a spatial resolution of 2m. The datasets are collected from three cities: XiangTan (XT), ChangSha (CS), and TuoKetuo (TK). The XT dataset consists of bitemporal phases: winter in 2018 and summer in 2019. Each phase of the XT dataset contains 4232 RGB RS image patches with a size of 512 × 512. The CS dataset consists of bitemporal phases: autumn in 2018 and summer in 2019. Each phase of the CS dataset contains 4232 RGB RS image patches with a size of 512 × 512. The TK dataset consists of bitemporal phases: spring in 2018 and summer in 2021. Each phase of the TK dataset contains 2882 RGB RS image patches with a size of 512 × 512. For XT and CS datasets, we annotated four land cover categories: farmland, forest, water, and artificial surface. The TK dataset includes the following four land cover categories: farmland, grassland, bare land, and artificial surface. We set up three temporal DA tasks to validate the effectiveness of the proposed framework. For example, we use XT dataset's 2018-winter images as the source domain and XT dataset's 2019-summer images as the target domain, denoted as XT 18-winter-to-19-summer. Similarly, the other two temporal DA tasks are denoted as CS 18-autumn-to-19-summer and TK 18-spring-to-21-summer. Some sample patches of three datasets are shown in Fig. 5. It can be seen that different land cover classes exhibit phenological heterogeneity, for example, evergreen forests exhibit varying degrees of greenness throughout the seasons, while farmlands are filled with green crops in summer but may be in a fallow period in winter or autumn. Meanwhile, compared with nonvegetated areas, vegetation is more susceptible to undergo visual differences with seasonal changes.

2) Implementation Details: All experiments are performed on a PC with 3.7 Hz Core E5-1620 CPU, GeForce GTX 1080Ti, and 24-GB RAM. We use DeepLab v3+ [7] with ResNet-50 as the semantic segmentation architecture. We set the initial learning rate LR = 0.002 for SGD optimizer, then adjust the learning rate by cosine annealing. The momentum is set to 0.9. We set $\lambda_1, \lambda_2, \lambda_3$, and λ_4 parameters in (28) to 1, 1, 0.5, and 0.5, respectively. For the XT 18-winter-to-19-summer task, confidence threshold value d is set as 0.7. For the CS 18-autumn-to-19-summer and TK 18-spring-to-21-summer tasks, confidence threshold d are set as 0.8. To fairly evaluate different DA methods, we use the same semantic segmentation architecture, optimizer, and hyperparameters of the optimizer.

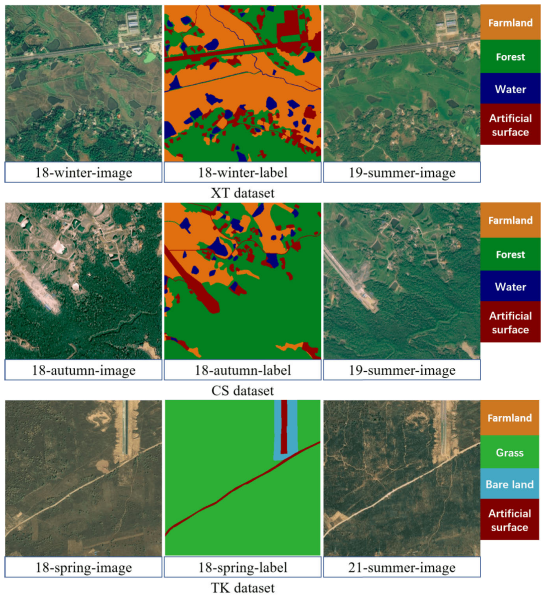


Fig. 5. Bitemporal images sampled from three datasets.

B. Comparison With State-of-the-Art DA Methods

To demonstrate the effectiveness of style transfer with semantic guidance for improving domain adaptive semantic segmentation, we compare the proposed SGST with state-of-the-art cross-domain semantic segmentation methods including one adversarial-learning-based, two self-training-based, and two style-transfer-based methods. AdvEnt [33] employs adversarial learning to learn domain-invariant features. CBST [52] utilizes high-confidence predictions as pseudolabels to further train the target classifier. ProDA [47] enhances the quality of pseudolabels by prototypical denoising. Both CycleGAN [51] and DAugNet [27] generate style-transferred images at a view of the global scene and use them to train the adaptation segmentation model. We assess the performance by the following evaluation criterion: the frequency weighted intersection over union (FWIoU), the overall accuracy (OA), the kappa coefficient, and the intersection over union (IoU) for each land cover class.

For the XT 18-winter-to-19-summer task, adaptation results are shown in Fig. 6. We can observe that without any adaptation strategy (Src Only), water and farmlands are prone to classification confusion in XT dataset. AdvEnt adopts global adversarial feature alignment but is insufficient to alleviate classification confusion among local areas. CBST

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

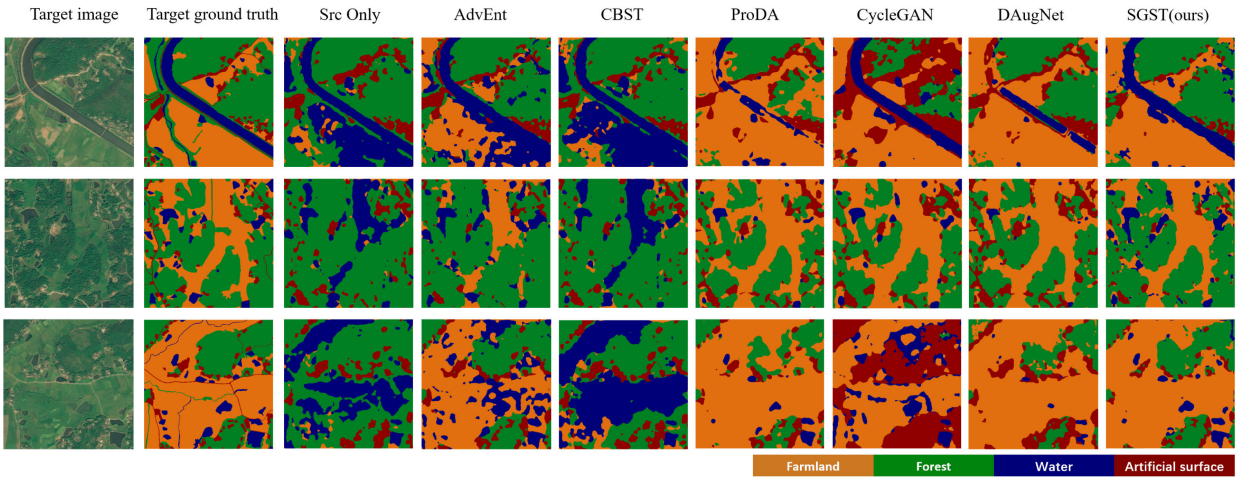


Fig. 6. Qualitative comparisons of adapted segmentation on XT 18-winter-to-19-summer.

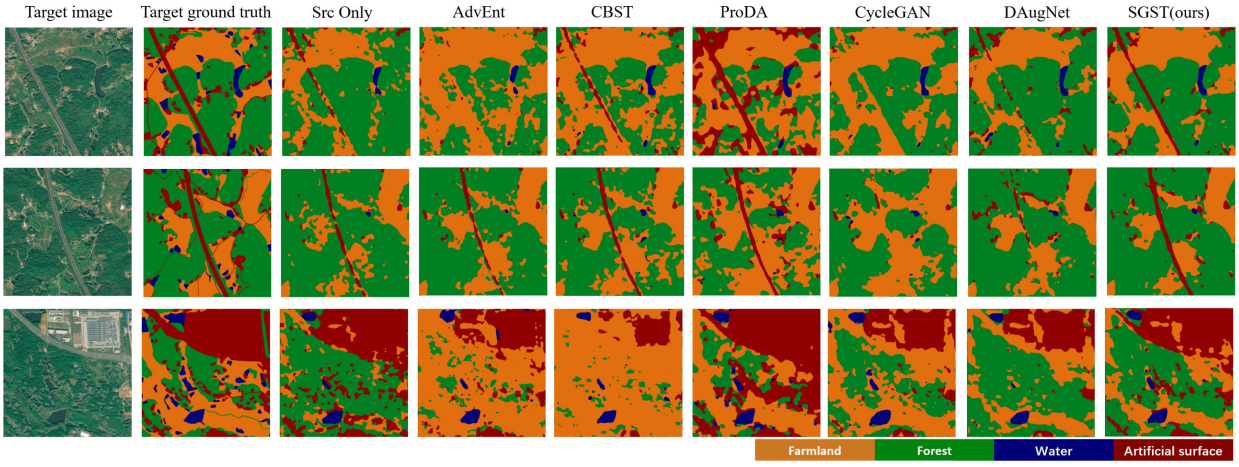


Fig. 7. Qualitative comparisons of adapted segmentation on CS 18-autumn-to-19-summer.

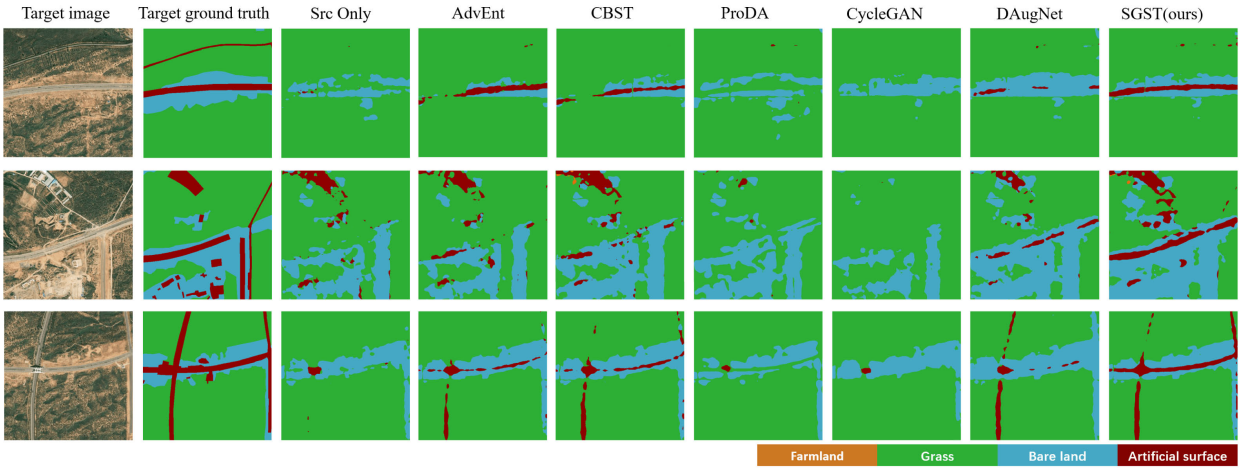


Fig. 8. Qualitative comparisons of adapted segmentation on TK 18-spring-to-21-summer.

adjusts the classifier minorly and also fails to address this issue. ProDA achieves improved results, yet it remains inadequate in achieving clear segmentation boundaries between water and farmlands. While CycleGAN and DAugNet reduce the classification confusion between water and farmlands, it increases classification confusion between other categories, such as forests and surrounding artificial surfaces. The adaptation results show that our framework efficiently reduces

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

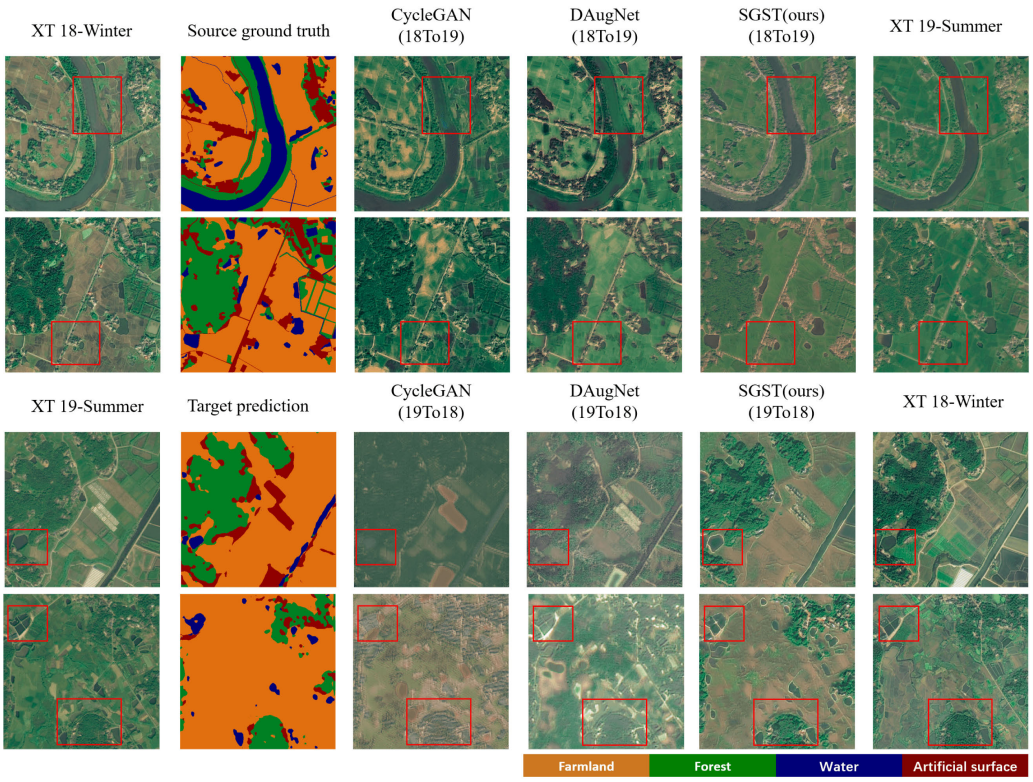


Fig. 9. Style transfer analysis between XT 18-winter and XT 19-summer.

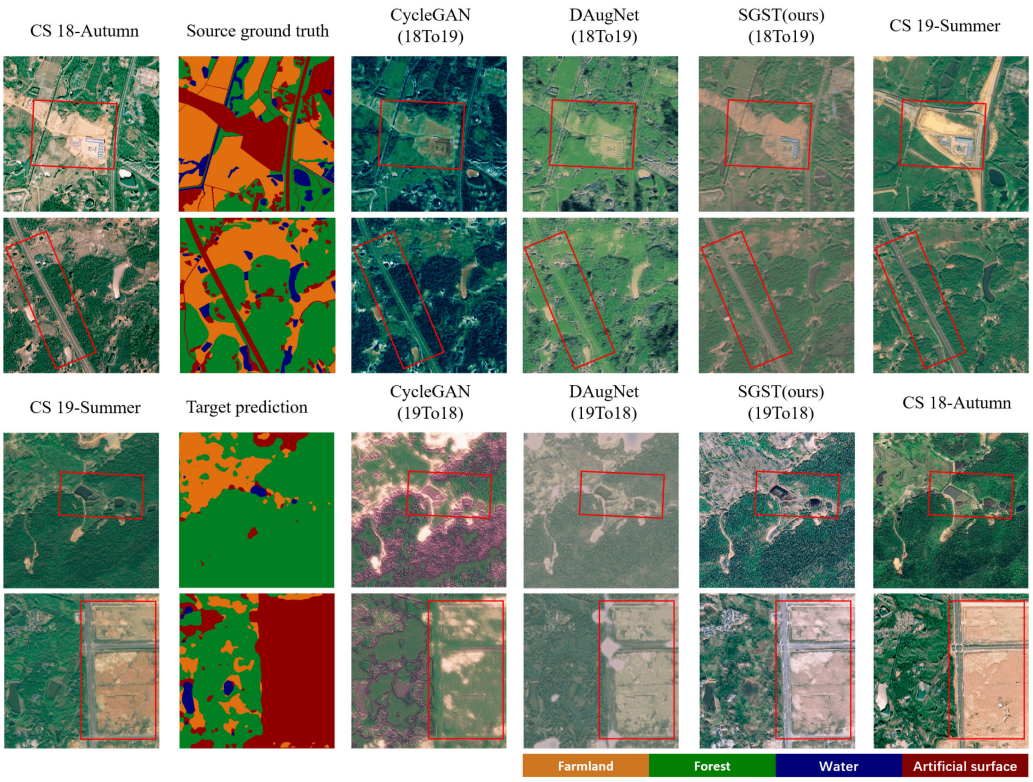


Fig. 10. Style transfer analysis between CS 18-autumn and CS 19-summer.

classification confusion and acquires more accurate segmentation boundaries, which can be ascribed to the style transfer with semantic guidance and the dual learning approach. The quantitative results are summarized in Table I. The proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

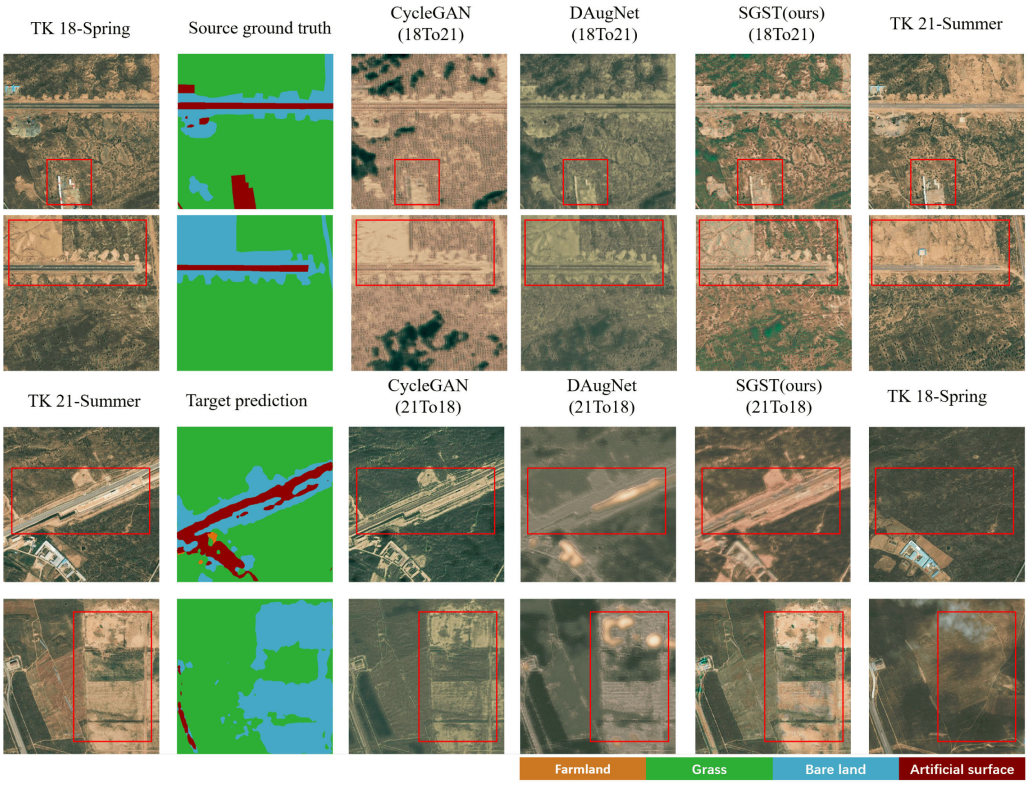


Fig. 11. Style transfer analysis between TK 18-spring and TK 21-summer.

TABLE I

COMPARISON RESULTS (%) OF XT 18-WINTER-TO-19-SUMMER IN TERMS OF IOU FOR EACH CLASS, OA, KAPPA, AND FWIOU

model	Farmland	Forest	Water	Artificial surface	OA	Kappa	FWIoU
Src Only	23.30	55.77	25.42	61.08	55.10	43.11	43.12
AdvEnt	53.22	66.03	41.54	63.56	71.08	61.30	57.05
CBST	22.95	66.50	27.41	63.72	61.96	50.49	47.27
ProDA	58.63	62.84	55.45	64.26	74.42	64.73	58.79
CycleGAN	57.94	64.51	39.23	62.68	71.02	61.19	57.43
DAugNet	62.64	63.37	36.57	62.11	73.61	63.50	57.70
SGST(ours)	63.94	68.62	60.77	66.52	77.25	68.91	63.53

TABLE II

COMPARISON RESULTS (%) OF CS 18-AUTUMN-TO-19-SUMMER IN TERMS OF IOU FOR EACH CLASS, OA, KAPPA, AND FWIOU

model	Farmland	Forest	Water	Artificial surface	OA	Kappa	FWIoU
Src Only	35.65	66.28	19.27	47.72	70.15	48.77	52.75
AdvEnt	41.70	70.79	17.65	52.61	72.22	56.39	57.58
CBST	40.83	67.76	21.64	52.89	70.58	53.81	55.81
ProDA	43.23	69.91	18.22	55.69	73.56	56.58	57.87
CycleGAN	41.48	67.82	21.04	49.62	71.11	52.94	55.19
DAugNet	41.95	69.72	24.03	50.53	73.09	54.38	56.56
SGST(ours)	48.15	75.74	19.87	57.12	76.88	62.23	62.26

SGST achieves superior performance, at least outperforming other adaptation methods by 5.71% in terms of FWIoU.

For the CS 18-autumn-to-19-summer task, adaptation results are shown in Fig. 7. We can also observe that though CycleGAN and DAugNet reduce classification confusion between farmlands and forests, artificial surfaces are prone to be misclassified as farmlands by their strategies. In contrast, the proposed SGST efficiently alleviates classification confusion with fewer adverse effects. This phenomenon demonstrates that the style transfer with semantic guidance is more effective in reducing temporal domain shift. Table II depicts the comparison of quantitative results. The proposed SGST outperform the nonadapted baseline by 9.41% and DAugNet by 5.6% in terms of FWIoU.

For the TK 18-spring-to-21-summer task, adaptation results are shown in Fig. 8. We make the same observation that the proposed SGST achieves more accurate segmentation results compared to the other compared DA methods. We list the quantitative results in Table III. The proposed SGST also achieves competitive performance.

TABLE III

COMPARISON RESULTS (%) OF TK 18-SPRING-TO-21-SUMMER IN TERMS OF IOU FOR EACH CLASS, OA, KAPPA, AND FWIOU

model	Farmland	Grass	Bare land	Artificial surface	OA	Kappa	FWIoU
Src Only	64.53	90.20	48.35	20.10	90.12	62.50	82.25
AdvEnt	71.93	91.32	50.28	36.54	91.27	67.15	84.05
CBST	63.27	91.59	56.97	32.11	91.10	69.43	84.38
ProDA	65.31	90.89	57.19	21.15	90.86	65.26	83.58
CycleGAN	70.84	90.76	56.28	3.88	90.66	63.68	82.75
DAugNet	69.11	92.16	59.23	25.78	91.63	70.14	84.87
SGST(ours)	67.21	92.85	63.93	39.98	92.56	73.81	86.42

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

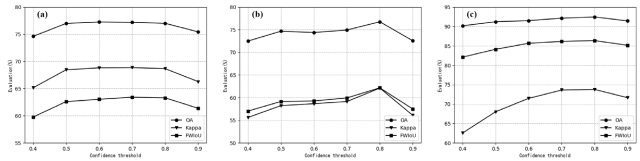


Fig. 12. Parameter analysis of the pseudosegmentation map threshold. (a) XT 18-winter-to-19-summer; (b) CS 18-autumn-to-19-summer; and (c) TK 18-spring-to-21-summer.

IV. Discussion

A. Effectiveness of Style Transfer With Semantic Guidance

In this section, we further discuss how the style transfer with semantic guidance is beneficial for adaptive segmentation of bitemporal RS images.

Fig. 9 displays style transfer results between XT 18-winter and XT 19-summer. We can see that CycleGAN and DAugNet pay more attention to learning the style of the majority land cover from the global scene rather than the minority. For example, in certain cases, pixels corresponding to water or artificial surfaces may be mistakenly transferred to the vegetation style due to the prevalence of farmlands and forests in the scene. Therefore, utilizing these global style-transferred images to train the adaptive segmentation model might exacerbate classification confusion, as illustrated in Fig. 6. In contrast, style-transferred images generated by the proposed SGST not only close to the visual gap of bitemporal RS images but exhibit the desired phenological heterogeneity among different land cover classes. This characteristic is advantageous for learning of features with domain-invariant and semantic consistency.

As depicted in Figs. 10 and 11, we can also observe that the proposed SGST effectively preserves the attributes of local objects in the RS scene. Nevertheless, CycleGAN and DAugNet have a tendency to excessively apply global scene style transfer, which could potentially compromise the attributes of local objects, particularly in areas with local semantic changes between bitemporal RS images. Therefore, the style transfer with semantic guidance is more advantageous in adaptive segmentation of bitemporal RS images.

B. Sensitivity Analysis of Parameters

In the dual learning approach, confidence threshold d guides the generation of target pseudosegmentation maps. To analyze the sensitivity of d, we evaluate the performance of our framework under different d on three temporal DA tasks. The results of different threshold d on three tasks are shown in Fig. 12. For the XT 18-winter-to-19-summer task, the best performance is attained when d is set as 0.7. For the other two tasks, we obtain the peak performance when d is set as 0.8. We can also observe that the performance drops when d is set too large (e.g. larger than 0.9). Larger d may generate fewer pseudosegmentation maps that are not enough to retrain the model, resulting in poor performance.

For (28), the weight λ_1, λ_2 is fixed to 1 and the sensitivity of λ_3 and λ_4 will be studied. In the XT 18-winter-to-19-summer task, as shown in Table IV, a small value for λ_3 is detrimental, while λ_3 between 0.1 and 1 is relatively stable. Regarding λ_4, as shown in Table IV, similar observations are obtained. Since $\mathcal{L}_{\text{cons}-s}$ and $\mathcal{L}_{\text{cons}-t}$ have already ignored the calculation in low-confidence prediction regions through threshold d, a small value for λ_3 and λ_4 cannot effectively utilize style-transferred images to align semantic features.

TABLE IV

Influence of λ_3 and λ_4. The Experiment Involves Retaining the Best Value of One of the Lambdas and Searching for the Best Value of the Other

λ_3	0.01	0.05	0.1	0.5	1
XT 18-To-19 FWIoU	61.37	62.45	63.29	63.46	63.41
λ_4	0.01	0.05	0.1	0.5	1
XT 18-To-19 FWIoU	61.76	62.59	63.34	63.53	63.48

TABLE V

Influence of Dual Learning on Adaptive Segmentation (%) of XT 18-Winter-to-19-Summer in Terms of IoU for Each Class, OA, Kappa, and FWIoU

model	Farmland	Forest	Water	Artificial surface	OA	Kappa	FWIoU
Src Only	23.30	55.77	25.42	61.08	55.10	43.11	43.12
Src+SCSN	61.72	66.41	48.69	62.63	74.67	65.58	60.76
Src+SCSN+DualL(m=2)	62.07	66.55	58.88	63.27	75.59	66.74	61.32
Src+SCSN+DualL(m=4)	63.94	68.62	60.77	66.52	77.25	68.91	63.53

TABLE VI

Influence of Dual Learning on Adaptive Segmentation (%) of CS 18-Autumn-to-19-Summer in Terms of IoU for Each Class, OA, Kappa, and FWIoU

model	Farmland	Forest	Water	Artificial surface	OA	Kappa	FWIoU
Src Only	35.65	66.28	19.27	47.72	70.15	48.77	52.75
Src+SCSN	41.69	70.61	19.86	51.16	73.21	54.61	56.83
Src+SCSN+DualL(m=2)	44.42	70.81	19.83	53.58	74.43	56.65	58.17
Src+SCSN+DualL(m=4)	48.15	75.74	19.87	57.12	76.88	62.23	62.26

TABLE VII

Influence of Dual Learning on Adaptive Segmentation (%) of TK 18-Spring-to-21-Summer in Terms of IoU for Each Class, OA, Kappa, and FWIoU

model	Farmland	Grass	Bare land	Artificial surface	OA	Kappa	FWIoU
Src Only	64.53	90.20	48.35	20.10	90.12	62.50	82.25
Src+SCSN	68.56	92.17	57.10	28.57	91.68	70.18	85.03
Src+SCSN+DualL(m=2)	68.74	92.39	58.81	35.73	92.00	71.55	85.51
Src+SCSN+DualL(m=4)	67.21	92.85	63.93	39.98	92.56	73.81	86.42

C. Effectiveness of Dual Learning Approach

Furthermore, we discussion the effectiveness of dual learning approach in the following two aspects.

First, dual learning aids the style transfer model in producing improved style-transferred images. As shown in Fig. 13, we present the style transfer results without dual learning, as well as the results with dual learning corresponding to $m = 2$ and $m = 4$. Without dual learning, the optimization of the style transfer model can only rely on crude pseudosegmentation maps generated in one iteration, resulting in relatively lower quality of style-transferred images. With the help of dual

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

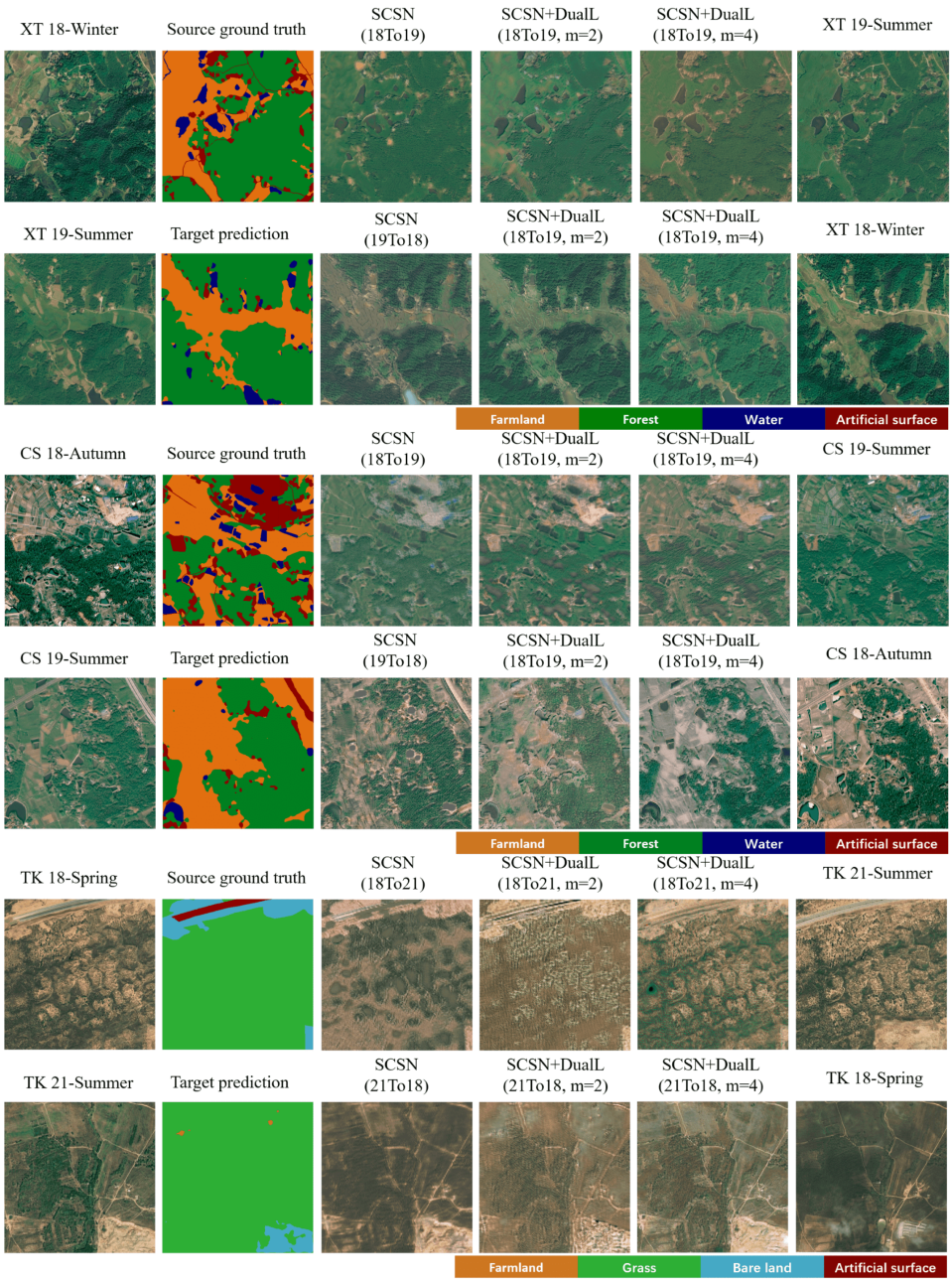


Fig. 13. Influence of dual learning on SGST.

TABLE VIII
EXECUTION COSTS OF MODELS

Model	Time complexity	Style transfer training time (s/epoch) in XT dataset
DAugNet	$O(n_{st} + n_{seg})$	1786
SGST(ours)	$O((n_{st} + n_{seg}) \cdot m)$	2101

* For each model, its style transfer training time is the sum of time used to train its style transfer model. The time complexity of the proposed model is $O((n_{st} + n_{seg}) \cdot m)$, where n_{st} is the number of style transfer training epochs, n_{seg} is the number of segmentation training epochs, and m is the dual learning iterations.

learning, the local object details in the style-transferred images become more realistic.

Second, dual learning contributes to the performance enhancement of the adaptive segmentation model. Tables V–VII show the quantitative results of adaptive segmentation without dual learning, as well as the results with dual learning corresponding to $m = 2$ and $m = 4$. As can be seen, the solution with dual learning achieves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

improved performance. It means the adaptive segmentation model and the style transfer model can be complementary to each other. Through dual learning, the adaptive segmentation model can achieve superior performance with the assistance of improved style-transferred images.

Table VIII reports the execution costs of models on an NVIDIA GeForce GTX 1080 Ti GPU. Compared with DAugNet, the SGST model has a higher time complexity due to its dual learning mechanism. However, in terms of the style transfer training process, the difference in time cost is very small.

V. Conclusion

This article first formulates the temporal domain shift problem as an issue of two-factor interference on feature distributions. To tackle this issue, we propose a SGST framework seamlessly integrating global feature alignment with local feature semantic matching. The proposed framework combines a style transfer model and an adaptive segmentation model into a dual learning manner. The basic idea is to first apply pseudosegmentation maps as semantic guidance to achieve style transfer at both global image and local semantic levels, and then make use of such style-transferred images to progressively obtain more accurate pseudosegmentation maps. Extensive experiments demonstrate that our proposed method is more effective for DA of bitemporal RS images compared with state-of-the-art methods.

As mentioned earlier, dual learning can enhance the performance of both the style transfer model and semantic segmentation model. However, this process is not trained end-to-end. Dual learning also results in longer training time for the proposed model. Moreover, this article only discusses the temporal domain shift of bitemporal RS images; how the style transfer techniques can better solve the temporal domain shift problem in long time series RS images remains an area for further exploration.

In the future, we plan to further improve the method in the following aspects: 1) redesign the dual learning process between style transfer model and semantic segmentation model to achieve end-to-end training; 2) explore the style transfer techniques under the temporal domain shift of long time series RS images; and 3) design text descriptions of global style variations and local style variations between different temporal phases, and introduce prior knowledge through large language models.

References

[1] J. Adrian, V. Sagan, and M. Maimaitijiang, "Sentinel SAR-optical fusion for crop type mapping using deep learning and Google Earth Engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 215–235, May 2021, doi: 10.1016/j.isprsjprs.2021.02.018.

[2] L. Bai, S. Du, X. Zhang, H. Wang, B. Liu, and S. Ouyang, "Domain adaptation for remote sensing image semantic segmentation: An integrated approach of contrastive learning and adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628313, doi: 10.1109/TGRS.2022.3198972.

[3] S. Bajocco, E. Raparelli, T. Teofili, M. Bascietto, and C. Ricotta, "Text mining in remotely sensed phenology studies: A review on research development, main topics, and emerging issues," *Remote Sens.*, vol. 11, no. 23, p. 2751, Nov. 2019, doi: 10.3390/rs11232751.

[4] E. F. Berra and R. Gaulton, "Remote sensing of temperate and boreal forest phenology: A review of progress, challenges and opportunities in the intercomparison of in-situ and satellite phenological metrics," *Forest Ecol. Manage.*, vol. 480, Jan. 2021, Art. no. 118663, doi: 10.1016/j.foreco.2020.118663.

[5] P. G. Brodrick, A. B. Davies, and G. P. Asner, "Uncovering ecological patterns with convolutional neural networks," *Trends Ecol. Evol.*, vol. 34, no. 8, pp. 734–745, Aug. 2019, doi: 10.1016/j.tree.2019.03.006.

[6] D. Brunner, G. Lemoine, and L. Bruzzone, "Earthquake damage assessment of buildings using VHR optical and SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 5, pp. 2403–2420, May 2010.

[7] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818, doi: 10.1007/978-3-030-01234-2_49.

[8] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 1–35, 2096.

[9] L. A. Gatys, A. S. Ecker, and M. Bethge, "Image style transfer using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2414–2423, doi: 10.1109/CVPR.2016.265.

[10] J. Ge, H. Tang, N. Yang, and Y. Hu, "Rapid identification of damaged buildings using incremental learning with transferred data from historical natural disaster cases," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 105–128, Jan. 2023, doi: 10.1016/j.isprsjprs.2022.11.010.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

[12] J. Hoffman et al., "CyCADA: Cycle-consistent adversarial domain adaptation," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 1989–1998.

[13] X. Huang and S. Belongie, "Arbitrary style transfer in real-time with adaptive instance normalization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1501–1510, doi: 10.1109/ICCV.2017.167.

[14] T. Kattenborn, J. Leitloff, F. Schiefer, and S. Hinz, "Review on convolutional neural networks (CNN) in vegetation remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 24–49, Mar. 2021, doi: 10.1016/j.isprsjprs.2020.12.010.

[15] T. Kim, M. Cha, H. Kim, J. K. Lee, and J. Kim, "Learning to discover cross-domain relations with generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1857–1865.

[16] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 778–782, May 2017, doi: 10.1109/LGRS.2017.2681128.

[17] G. Li, G. Kang, W. Liu, Y. Wei, and Y. Yang, "Content-consistent matching for domain adaptive semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 440–456, doi: 10.1007/978-3-030-58568-6_26.

[18] S. Li, F. Lv, B. Xie, C. H. Liu, J. Liang, and C. Qin, "Bi-classifier determinacy maximization for unsupervised domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 8455–8464, doi: 10.1609/aaai.v35i10.17027.

[19] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021, doi: 10.1016/j.isprsjprs.2021.07.007.

[20] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021, doi: 10.1016/j.isprsjprs.2021.02.009.

[21] Q. Lian, F. Lv, L. Duan, and B. Gong, "Constructing self-motivated pyramid curriculums for cross-domain semantic segmentation: A non-adversarial approach," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 6758–6767.

[22] G. Lu, Z. Zhou, Y. Song, K. Ren, and Y. Yu, "Guiding the one-to-one mapping in CycleGAN via optimal transport," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 4432–4439, doi: 10.1609/aaai.v33i01.33014432.

[23] Y. Luo, L. Zheng, T. Guan, J. Yu, and Y. Yang, "Taking a closer look at domain shift: Category-level adversaries for semantics consistent domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2507–2516, doi: 10.1109/CVPR.2019.00261.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.

[24] S. Saha, F. Bovolo, and L. Bruzzone, "Building change detection in VHR SAR images via unsupervised deep transcoding," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1917–1929, Mar. 2021, doi: 10.1109/TGRS.2020.3000296.

[25] M. N. Subhani and M. Ali, "Learning from scale-invariant examples for domain adaptation in semantic segmentation," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, Aug. 2020, pp. 290–306, doi: 10.1007/978-3-030-58542-6_18.

[26] C. Tao, J. Qi, M. Guo, Q. Zhu, and H. Li, "Self-supervised remote sensing feature learning: Learning paradigms, challenges, and future works," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610426, doi: 10.1109/TGRS.2023.3276853.

[27] O. Tasar, A. Giros, Y. Tarabalka, P. Alliez, and S. Clerc, "DAugNet: Unsupervised, multisource, multitarget, and life-long domain adaptation for semantic segmentation of satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1067–1081, Feb. 2021, doi: 10.1109/TGRS.2020.3006161.

[28] W. Tranheden, V. Olsson, J. Pinto, and L. Svensson, "DACS: Domain adaptation via cross-domain mixed sampling," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Jan. 2021, pp. 1379–1389, doi: 10.1109/WACV48630.2021.00142.

[29] Y. Tsai, W. Hung, S. Schulter, K. Sohn, M. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7472–7481, doi: 10.1109/CVPR.2018.00780.

[30] D. Tuia, C. Persello, and L. Bruzzone, "Domain adaptation for the classification of remote sensing data: An overview of recent advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 41–57, Jun. 2016, doi: 10.1109/MGRS.2016.2548504.

[31] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Improved texture networks: Maximizing quality and diversity in feed-forward stylization and texture synthesis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6924–6932, doi: 10.1109/CVPR.2017.437.

[32] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. P. Pérez, "DADA: Depth-aware domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 7364–7373, doi: 10.1109/ICCV.2019.00746.

[33] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Pérez, "Advent: Adversarial entropy minimization for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 2517–2526, doi: 10.1109/CVPR.2019.00262.

[34] H. Wang, C. Tao, J. Qi, R. Xiao, and H. Li, "Avoiding negative transfer for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4413215, doi: 10.1109/TGRS.2022.3201688.

[35] J. Wang, A. Ma, Y. Zhong, Z. Zheng, and L. Zhang, "Cross-sensor domain adaptation for high spatial resolution urban land-cover mapping: From airborne to spaceborne imagery," *Remote Sens. Environ.*, vol. 277, Aug. 2022, Art. no. 113058, doi: 10.1016/j.rse.2022.113058.

[36] Z. Wang et al., "Alleviating semantic-level shift: A semi-supervised domain adaptation method for semantic segmentation," in *Proc. CVPR Workshops*, Jun. 2020, pp. 936–937, doi: 10.1109/CVPRW50498.2020.00476.

[37] Z. Wang et al., "Differential treatment for stuff and things: A simple unsupervised domain adaptation method for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12632–12641, doi: 10.1109/CVPR42600.2020.01265.

[38] Z. Wang, H. Zhang, W. He, and L. Zhang, "Cross-phenological-region crop mapping framework using Sentinel-2 time series imagery: A new perspective for winter crops in China," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 200–215, Nov. 2022, doi: 10.1016/j.isprsjprs.2022.09.010.

[39] Z. Wei, X. Gu, Q. Sun, X. Hu, and Y. Gao, "Analysis of the spatial and temporal pattern of changes in abandoned farmland based on long time series of remote sensing data," *Remote Sens.*, vol. 13, no. 13, p. 2549, Jun. 2021, doi: 10.3390/rs13132549.

[40] D. Wittich and F. Rottensteiner, "Appearance based deep domain adaptation for the classification of aerial images," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 82–102, Oct. 2021, doi: 10.1016/j.isprsjprs.2021.08.004.

[41] X. Wu, Z. Wu, H. Guo, L. Ju, and S. Wang, "DANNet: A one-stage domain adaptation network for unsupervised nighttime semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15769–15778, doi: 10.1109/CVPR46437.2021.01551.

[42] Q. Xu, X. Yuan, and C. Ouyang, "Class-aware domain adaptation for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4500317, doi: 10.1109/TGRS.2020.3031926.

[43] L. Yan, B. Fan, H. Liu, C. Huo, S. Xiang, and C. Pan, "Triplet adversarial domain adaptation for pixel-level classification of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3558–3573, May 2020, doi: 10.1109/TGRS.2019.2958123.

[44] Z. Yi, H. Zhang, P. Tan, and M. Gong, "DualGAN: Unsupervised dual learning for image-to-image translation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2849–2857, doi: 10.1109/ICCV.2017.310.

[45] B. Yuan, D. Zhao, S. Shao, Z. Yuan, and C. Wang, "Birds of a feather flock together: Category-divergence guidance for domain adaptive segmentation," *IEEE Trans. Image Process.*, vol. 31, pp. 2878–2892, 2022, doi: 10.1109/TIP.2022.3162471.

[46] C. Zhang et al., "Joint deep learning for land cover and land use classification," *Remote Sens. Environ.*, vol. 221, pp. 173–187, Feb. 2019, doi: 10.1016/j.rse.2018.11.014.

[47] P. Zhang, B. Zhang, T. Zhang, D. Chen, Y. Wang, and F. Wen, "Prototypical pseudo label denoising and target structure learning for domain adaptive semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12414–12424.

[48] Y. Zhang, Z. Qiu, T. Yao, D. Liu, and T. Mei, "Fully convolutional adaptation networks for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6810–6818, doi: 10.1109/CVPR.2018.00712.

[49] Y. Zhang, X. Li, J. Huang, H. Guan, and H. Huang, "A novel framework for urban land cover change detection with NASA's black marble nighttime lights product," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411009.

[50] J. Zhu, Y. Guo, G. Sun, L. Yang, M. Deng, and J. Chen, "Unsupervised domain adaptation semantic segmentation of high-resolution remote sensing imagery with invariant domain-level prototype memory," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603518, doi: 10.1109/TGRS.2023.3243042.

[51] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232, doi: 10.1109/ICCV.2017.244.

[52] Y. Zou, Z. Yu, B. V. Kumar, and J. Wang, "Unsupervised domain adaptation for semantic segmentation via class-balanced self-training," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 289–305, doi: 10.1007/978-3-030-01219-9_18.

[53] Y. Zou, Z. Yu, X. Liu, B. Kumar, and J. Wang, "Confidence regularized self-training," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 5982–5991, doi: 10.1109/ICCV.2019.00608.



Hao Wang received the B.S. degree from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2018, where he is currently pursuing the Ph.D. degree.

His research interests include remote sensing and computer vision, deep learning, transfer learning, and their applications in remote sensing.



Mingning Guo received the B.E. degree from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2022, where he is currently pursuing the Ph.D. degree.

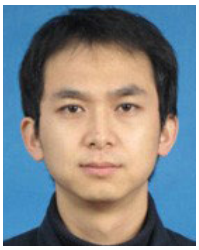
His research interests include remote sensing and computer vision, machine learning, deep learning, and their applications in remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.



Shaoxian Li received the B.S. degree from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2024, where he is currently pursuing the Ph.D. degree.

His research interests include remote sensing, deep learning, 3-D generation, embodied intelligence, and their applications in remote sensing.



Haifeng Li (Member, IEEE) received the master's degree in transportation engineering from the South China University of Technology, Guangzhou, China, in 2005, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2009.

He was a Research Associate with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong, in 2011, and a Visiting Scholar with the University of Illinois at Urbana-Champaign, Urbana, IL, USA, from 2013 to 2014. He is currently a Professor with the School of Geosciences and Info-Physics, Central South University, Changsha, China. He has authored over 30 journal articles. He is a reviewer for many journals. His research interests include geo/remote sensing big data, machine/deep learning, and artificial/brain-inspired intelligence.



Chao Tao (Member, IEEE) received the B.S. degree from the School of Mathematics and Statistics, Huazhong University of Science and Technology, Wuhan, China, in 2007, and the Ph.D. degree from the Institution of Patter Recognition and Artificial Intelligence, Huazhong University of Science and Technology, Wuhan, in 2012.

He is currently a Processor with the School of Geosciences and Info-Physics, Central South University, Changsha, China. He is also a Communication Evaluation Expert for the National Natural Science Foundation of China. He has authored more than 30 peer-reviewed articles in international journals from multiple domains such as remote sensing and computer vision. He has been frequently serving as a reviewer for several international journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), *Photogrammetric Engineering and Remote Sensing* (PERS), and *ISPRS Journal of Photogrammetry and Remote Sensing* (ISPRS–JPRS). His research interests include computer vision, machine learning, deep learning, and their applications in remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:42:37 UTC from IEEE Xplore. Restrictions apply.